

The report that got slower every month

The same product, eighteen months on. Two hundred thousand users, two million tasks.

WHAT HAPPENED

Every Monday morning an internal report goes out to the customer success team: every user, how many tasks they are carrying, when they last signed up. It is used to spot accounts that have gone quiet.

At launch it took under a second. Nobody thought about it again.

It now takes so long that the scheduled job is killed before it finishes. Customer success has not had the report in three weeks and has started asking for it by hand.

Two engineers have already looked at it. One **added an index on the created_at column**; it made no measurable difference. The other **added four more indexes** and reported that writes across the product got noticeably slower, so they were rolled back.

Nothing about the report has changed since it was written. Only the amount of data has.

WHAT YOU HAVE BEEN TOLD

- The report is one page of SQL — six queries, run in sequence.
- The tasks table has grown from about 40,000 rows to **2,000,000**.
- **An index on created_at already exists** and the report still does not use it.
- One of the six queries is for a search box the team added last year. It has always been slow and everyone has quietly accepted it.
- Adding indexes was tried and made things worse elsewhere.
- You have the schema, the row counts, and all six queries.

YOUR FIRST HOUR

- 1 Why did adding an index on created_at change nothing?
- 2 Adding four indexes made writes slower. Was that the wrong instinct, or the wrong indexes?
- 3 How would you decide which queries to fix first?
- 4 One of the six cannot be fixed by rewriting it. What do you tell the product manager?
- 5 What would you measure before changing anything?